SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
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॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A++' Grade

CSE - Academic Year 2026-27
Java Programming Lab | Batch 2025-29

ASSIGNMENT : 18

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Batch	CSE A - 2
Class	Java Programming Lab
Academic Year & Semester	2025-29 : Semester 3
Date of Submission	

TITLE : Write a Java GUI program to demonstrate button click event handling.

Learning Objectives : To implement button click events using Java event handling mechanisms.

Theory :

Event handling allows Java programs to respond to user actions such as button clicks, mouse movements, and keyboard input. Java Swing uses the **ActionListener** interface to handle button click events. When a user clicks a button, the **actionPerformed()** method is executed.

Event-driven programming makes applications interactive by connecting user actions with program functionality. It is commonly used in graphical user interface applications.

Exercise :

1. Create a GUI calculator where buttons perform addition and subtraction.

```
import javax.swing.*;
import java.awt.event.*;

public class java18a {
    public static void main(String[] args) {

        JFrame frame = new JFrame("Simple Calculator");

        JLabel label1 = new JLabel("First Number:");
        JLabel label2 = new JLabel("Second Number:");

        JTextField field1 = new JTextField();
        JTextField field2 = new JTextField();

        JButton addButton = new JButton("Add");
        JButton subButton = new JButton("Subtract");

        JLabel result = new JLabel("Result: ");

        label1.setBounds(40, 30, 100, 30);
        field1.setBounds(150, 30, 150, 30);

        label2.setBounds(40, 70, 100, 30);
        field2.setBounds(150, 70, 150, 30);

        addButton.setBounds(70, 120, 100, 30);
        subButton.setBounds(180, 120, 100, 30);

        result.setBounds(100, 170, 200, 30);

        addButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                int a = Integer.parseInt(field1.getText());
                int b = Integer.parseInt(field2.getText());
```

```
                result.setText("Result: " + (a + b));
            }
        });

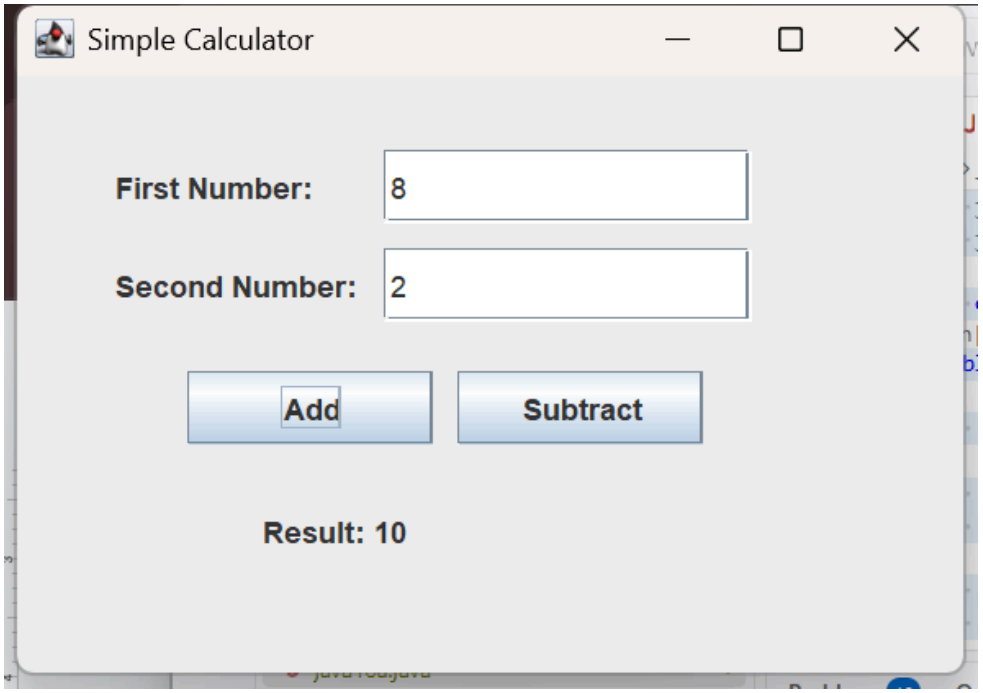
        subButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                int a = Integer.parseInt(field1.getText());
                int b = Integer.parseInt(field2.getText());

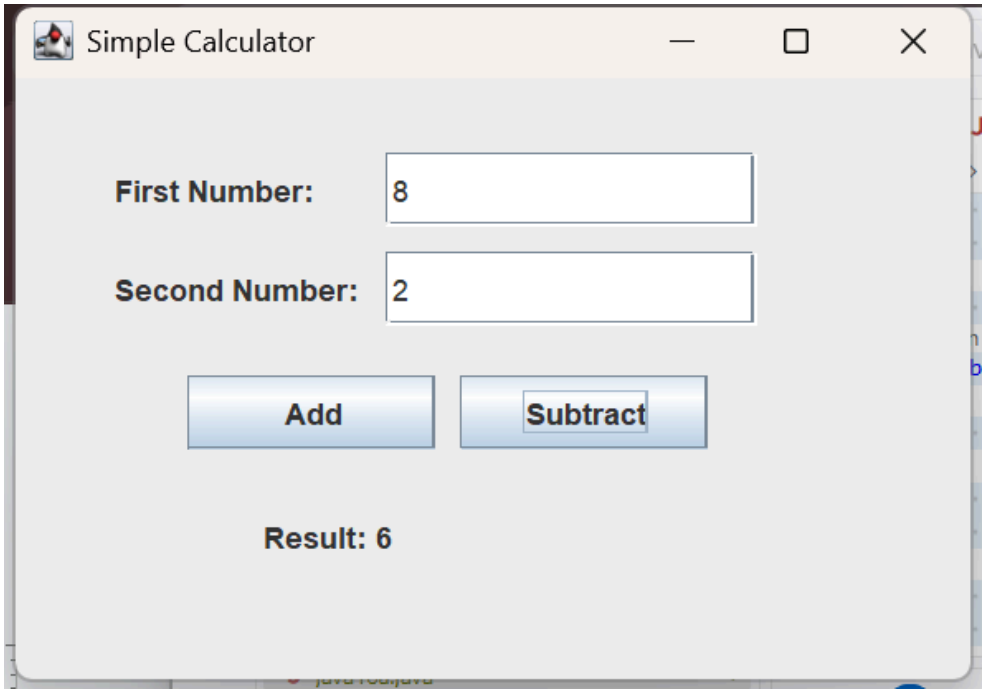
                result.setText("Result: " + (a - b));
            }
        });

        frame.add(label1);
        frame.add(field1);
        frame.add(label2);
        frame.add(field2);
        frame.add(addButton);
        frame.add(subButton);
        frame.add(result);

        frame.setSize(400, 280);
        frame.setLayout(null);
        frame.setVisible(true);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```

OUTPUT :


Simple Calculator
First Number:
8
Second Number:
2
Add
Subtract
Result: 10


Simple Calculator
First Number:
8
Second Number:
2
Add
Subtract
Result: 6

2. Create a Bank Balance Calculator GUI application using Java Swing where the user enters the initial balance and transaction amount. Use buttons to perform addition (deposit) and subtraction (withdrawal) and display the updated balance.

```
import java.awt.event.*;
import javax.swing.*;

public class BankBalanceCalculator implements ActionListener {

    JFrame frame;
    JTextField balanceField, amountField, resultField;
    JButton depositButton, withdrawButton;

    BankBalanceCalculator() {

        frame = new JFrame("Bank Balance Calculator");

        JLabel balanceLabel = new JLabel("Initial Balance:");
        JLabel amountLabel = new JLabel("Transaction Amount:");
        JLabel resultLabel = new JLabel("Updated Balance:");

        balanceField = new JTextField();
        amountField = new JTextField();
        resultField = new JTextField();

        resultField.setEditable(false);

        depositButton = new JButton("Deposit");
        withdrawButton = new JButton("Withdraw");

        balanceLabel.setBounds(40, 40, 130, 30);
        balanceField.setBounds(180, 40, 180, 30);

        amountLabel.setBounds(40, 80, 130, 30);
        amountField.setBounds(180, 80, 180, 30);

        resultLabel.setBounds(40, 120, 130, 30);
        resultField.setBounds(180, 120, 180, 30);
```

```
        depositButton.setBounds(80, 170, 110, 35);
        withdrawButton.setBounds(220, 170, 110, 35);

        frame.add(balanceLabel);
        frame.add(balanceField);

        frame.add(amountLabel);
        frame.add(amountField);

        frame.add(resultLabel);
        frame.add(resultField);

        frame.add(depositButton);
        frame.add(withdrawButton);

        depositButton.addActionListener(this);
        withdrawButton.addActionListener(this);

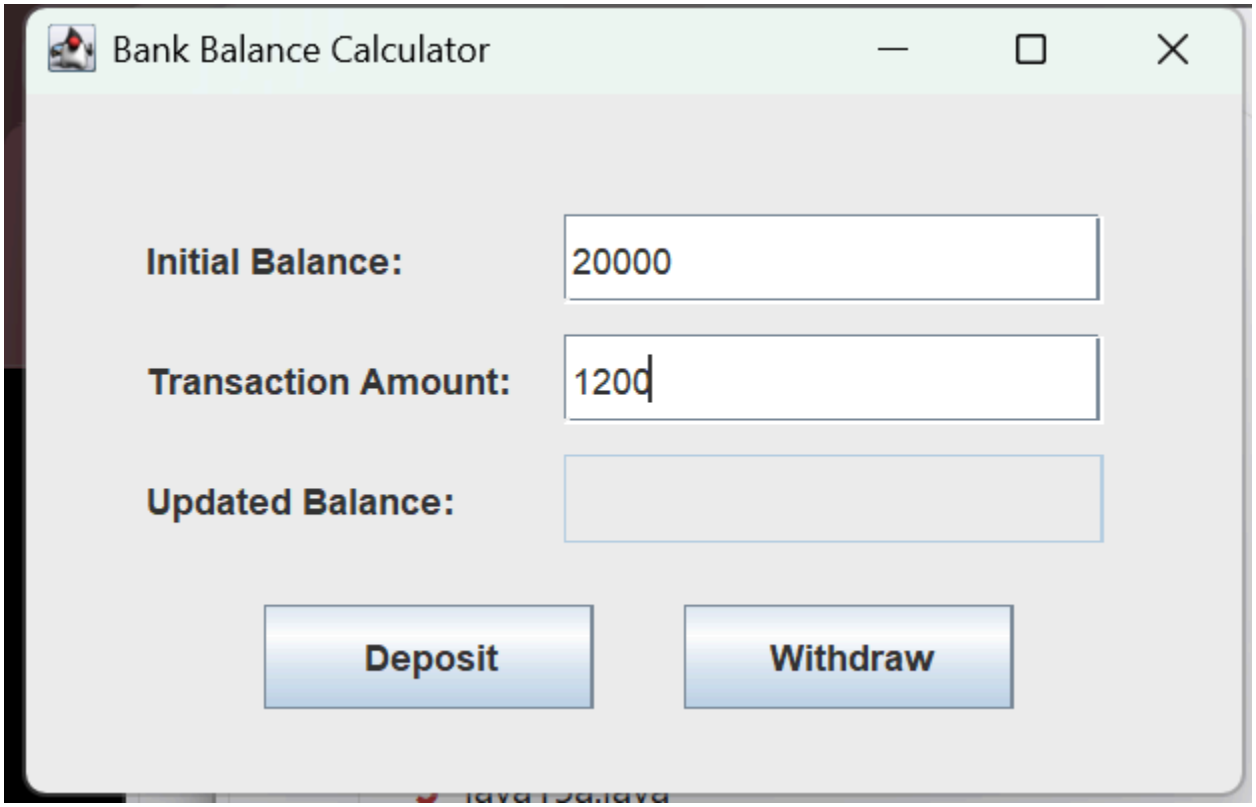
        frame.setSize(420, 270);
        frame.setLayout(null);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }

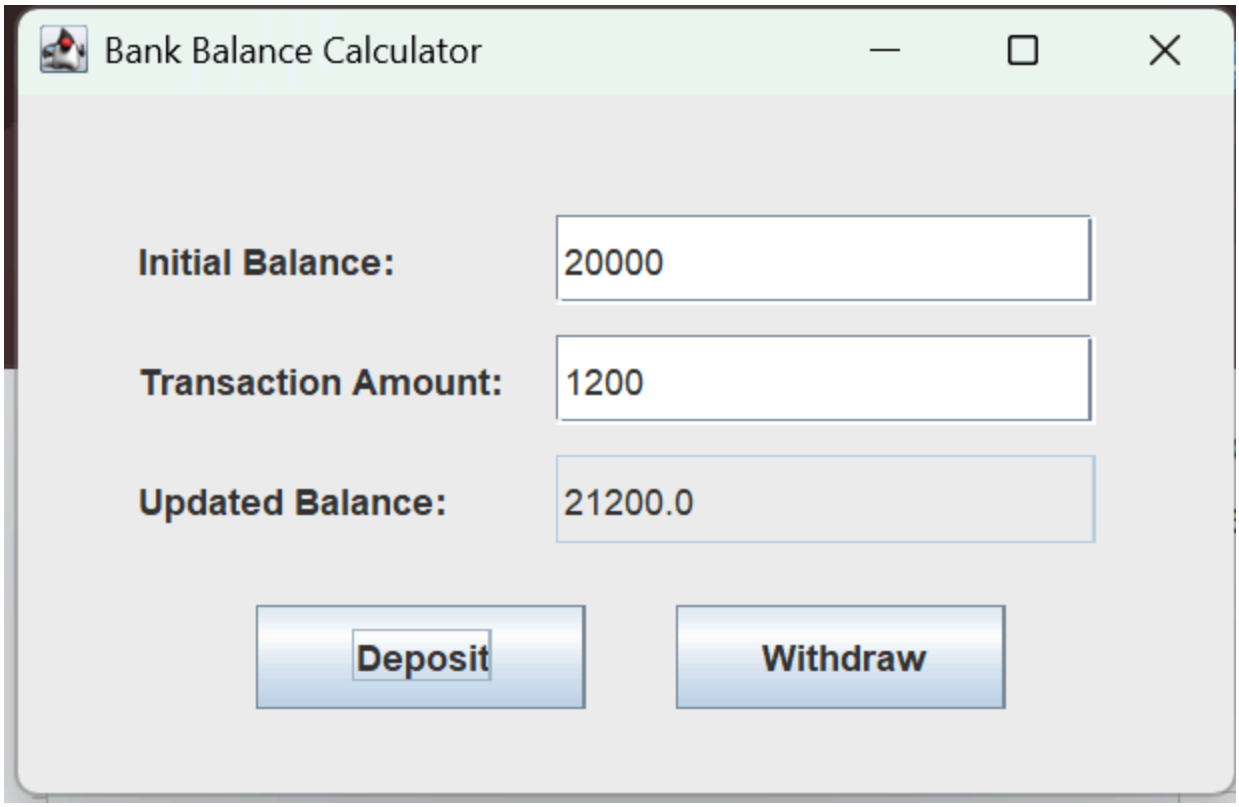
    public void actionPerformed(ActionEvent e) {

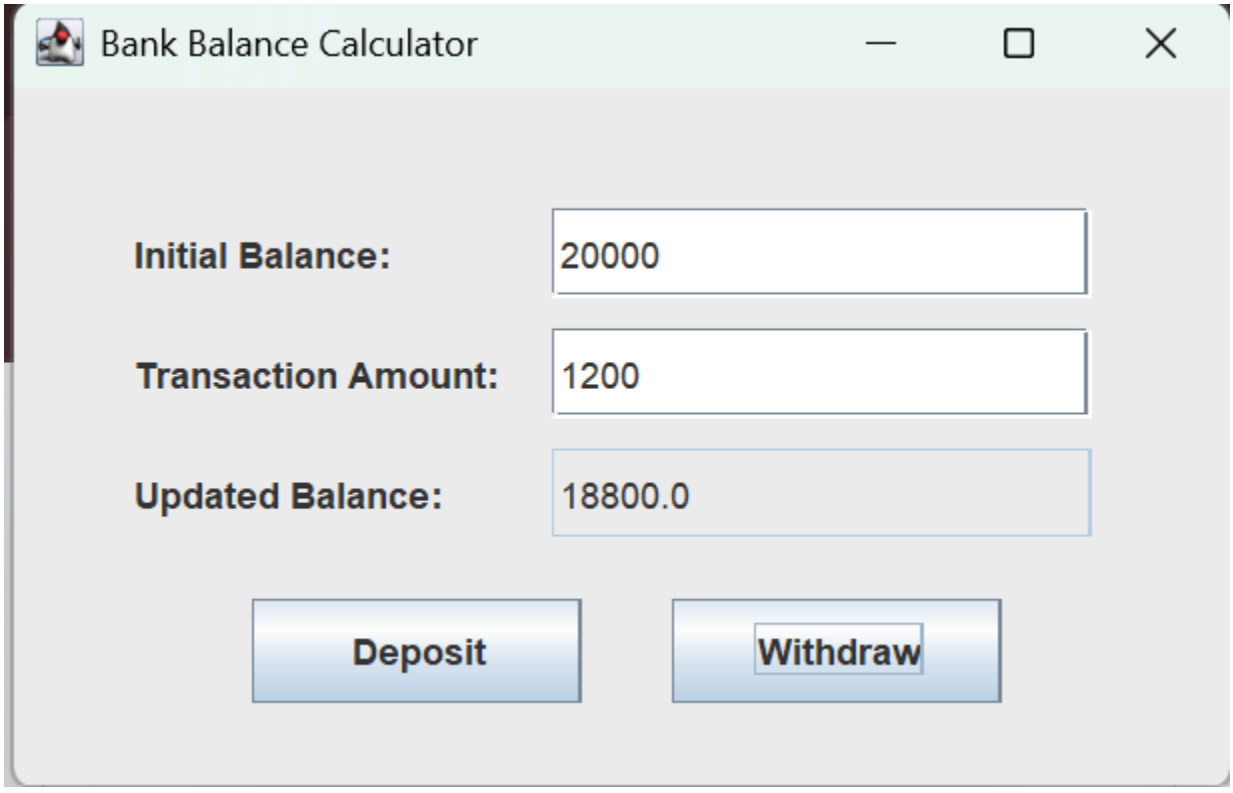
        double balance = Double.parseDouble(balanceField.getText());
        double amount = Double.parseDouble(amountField.getText());
        if (e.getSource() == depositButton) {
            balance = balance + amount;
        }
        else if (e.getSource() == withdrawButton) {
            balance = balance - amount;
        }
        resultField.setText(String.valueOf(balance));
    }

    public static void main(String[] args) {
        new BankBalanceCalculator();
    }
}
```

OUTPUT :


Bank Balance Calculator
Initial Balance:
20000
Transaction Amount:
1200
Updated Balance:
Deposit
Withdraw


Bank Balance Calculator
Initial Balance:
20000
Transaction Amount:
1200
Updated Balance:
21200.0
Deposit
Withdraw


Bank Balance Calculator
Initial Balance:
20000
Transaction Amount:
1200
Updated Balance:
18800.0
Deposit
Withdraw

Outcome :

- Understood button click event handling using Java Swing and ActionListener.
- Implemented interactive GUI applications where buttons perform specific operations.
- Learned how user actions can trigger calculations and update displayed results.

Conclusion :

This assignment demonstrated button click event handling in Java using Swing and the **ActionListener** interface. The calculator and bank balance applications successfully perform addition and subtraction operations and display the updated results based on user actions. This demonstrates how event handling makes Java GUI applications interactive and user-friendly.